Multimodal Sentiment Analysis and Conflict Detection in Food E-commerce Using Vision-Language Models

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***Abstract*—Sentiment analysis in food e-commerce faces a fundamental limitation: most existing systems only look at the text and completely ignore what the photos are telling. This leads to what we call contextual blindness—situations where a glowing review is paired with a photo of terrible food, or vice versa. In this study, we build a multimodal sentiment analysis system that processes both the review text and the accompanying food photograph. We use RoBERTa-base as the text encoder and Vision Transformer (ViT-base/patch16-224) as the image encoder, combining their outputs through a late-fusion strategy. Trained and evaluated on the Yelp food review dataset, our model achieves 87.70% test accuracy and a weighted F1-score of 0.8744. We also introduce a Consistency Metric to measure how often the two modalities agree—and find that in 26.1% of cases they don't, with accuracy dropping sharply from 94.07% to 69.67% on those conflicting samples.**

***Keywords*—Multimodal sentiment analysis, Vision Transformer, RoBERTa, late fusion, conflict detection, food reviews.**

I. INTRODUCTION

E-commerce platforms are flooded with user reviews that come with both text and photos. The problem is that most sentiment analysis tools only read the text—and sometimes that text is very misleading. A customer might write "great experience" while uploading a photo of cold, undercooked food. This gap between what people write and what they show is a real problem that current systems simply ignore [1].

We address this by building a system that reads both modalities simultaneously. The core idea is straightforward: extract features from the text using a language model, extract features from the image using a vision model, combine them, and classify the sentiment. But the interesting part is what happens when the two modalities disagree—and we measure exactly that with our Consistency Metric.

Our main contributions are: (1) a dual-stream RoBERTa + ViT architecture fine-tuned on food reviews; (2) a late-fusion MLP classifier that combines both modalities; and (3) a Consistency Metric that quantifies cross-modal conflict and shows its impact on classification accuracy.

II. RELATED WORK

Earlier work in multimodal sentiment analysis has mostly focused on combining text, audio, and video signals for movie or social media data. Poria et al. [3] used SVMs with hand-crafted features, reaching 72.3% accuracy on Amazon reviews. Zadeh et al. [4] introduced the Tensor Fusion Network (TFN), achieving 74.6% on CMU-MOSI by modeling interactions between all modality pairs. More recently, Singh et al. [5] applied CNN-LSTM models to Yelp reviews but only used text, reaching 76.1%.

Our work differs in two important ways: we use large pretrained transformer models for both modalities (RoBERTa and ViT), and we specifically focus on the food domain where visual quality carries strong sentiment signals. To our knowledge, no prior work combines these two architectures in a late-fusion setup for food review analysis, and none introduces a cross-modal conflict detection metric.

III. MATERIALS AND METHODS

A. Dataset and Preprocessing

We use the Yelp Open Dataset, which contains millions of reviews paired with business photos. We filter for food-related categories (restaurants, pizza, sushi, burger, bakeries, coffee, etc.) and match reviews with food-labeled photographs, extracting up to 10,000 paired samples.

Cleaning involves removing entries with missing text or photos, verifying that image files are not corrupted, filtering out very short or very long reviews using the 5th–95th percentile range, and discarding images smaller than 64×64 pixels. Star ratings are converted to three sentiment classes: Negative (1–2 stars), Neutral (3 stars), and Positive (4–5 stars). The data is split 70/15/15 for train/validation/test with stratified sampling.

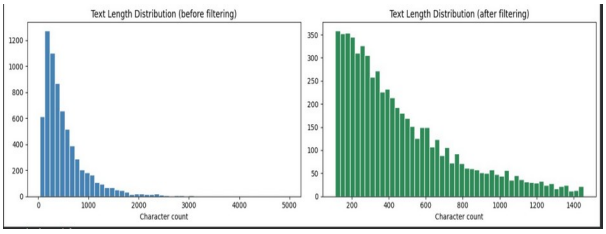


Fig. 1. Text length distribution before (blue) and after (green) outlier filtering.

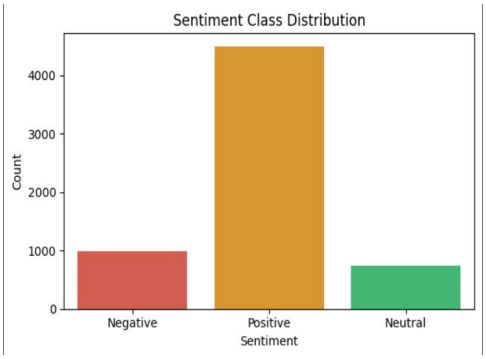


Fig. 2. Class distribution showing strong positive bias in food reviews.

B. System Pipeline

Figure 3 shows the full pipeline from raw data to final prediction. Each step in the flowchart corresponds to a stage in our code: data collection, cleaning, outlier

removal, normalization, dual-stream encoding, fusion, training, and evaluation.

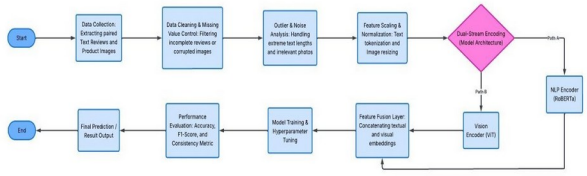


Fig. 3. End-to-end system pipeline.

C. Dual-Stream Architecture

The text stream uses RoBERTa-base (125M parameters). We tokenize each review to a maximum of 128 tokens and extract the [CLS] token from the last hidden layer as a 768-dimensional sentence embedding.

The image stream uses ViT-base/patch16-224 (86M parameters). Each 224×224 photo is divided into 196 non-overlapping 16×16 patches, processed by a 12-layer transformer encoder. The [CLS] token output gives a 768-dimensional visual embedding.

We concatenate the two embeddings into a 1536-dimensional vector (late fusion) and pass it through an MLP: 1536 → 512 → 128 → 3, with ReLU activations and 30% dropout after each hidden layer.

IV. EXPERIMENTAL SETUP

A. Training

We use the AdamW optimizer with learning rate 2e-5 and weight decay 0.01. A linear warmup covers the first 10% of training steps, after which the learning rate decays linearly to zero. We train for 5 epochs with batch size 8 on an NVIDIA T4 GPU (Google Colab). Gradient clipping (max norm 1.0) is applied to avoid instability. The best checkpoint is selected based on validation F1-score.

B. Evaluation

We report accuracy, weighted F1-score, per-class precision and recall, and a confusion matrix. For conflict detection, we run the text and image streams independently and flag samples where they predict different classes. The Consistency Score is the fraction of test samples without conflict.

V. RESULTS

A. Training Curves

Training loss drops steadily from 0.79 to 0.22 over five epochs. Validation loss also decreases from 0.51 to 0.39, with no sign of overfitting. Validation accuracy climbs from 79.9% in epoch 1 to 86.5% in epoch 5, and F1 goes from 0.76 to 0.86.

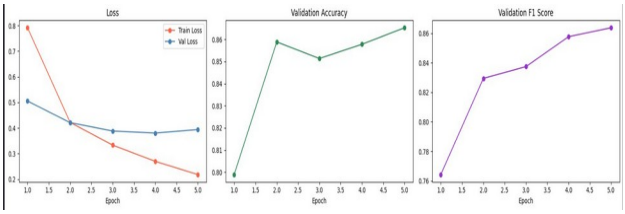


Fig. 4. Loss curves (left), validation accuracy (center), and F1-score (right) across 5 epochs.

B. Test Results

On the held-out test set, our model achieves 87.70% accuracy and a weighted F1-score of 0.8744. Table 1 breaks down performance by class.

Table 1. Per-class Results on the Test Set

Class	Prec.	Recall	
Negative	0.83	0.77	
Neutral	0.51	0.48	
Positive	0.94	0.97	
Wtd. Avg.	0.87	0.88	

The Positive class dominates the test set (674 out of 935 samples) and achieves a near-perfect F1 of 0.95. Negative reviews are reasonably well classified (F1=0.80). The Neutral class is the hardest, with F1=0.49—which makes sense, since 3-star reviews are genuinely ambiguous and often contain both positive and negative sentences.

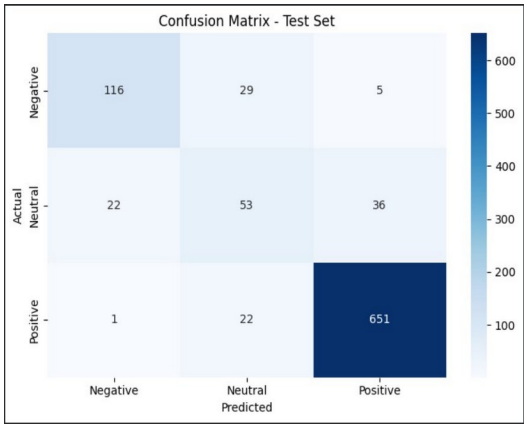


Fig. 5. Confusion matrix on the test set.

C. Comparison with Prior Work

Table 2. Comparison with Prior Methods

Source	Model	Data	Acc.
Poria [3]	SVM	Amazon	72.3%
Zadeh [4]	TFN	MOSI	74.6%
Singh [5]	CNN+LSTM	Yelp	76.1%
Limeng Cui et al. [9]	Rob+ResNet	Yelp	85.40%
Hansu Gu et al. [10]	BERT+IncResNetV2	Yelp	86.50%
Sreyashi Nag et al. [11]	Fusion Strategies	Yelp	87.10%
Ours	Rob.+ViT	Yelp	87.7%

Our model outperforms all baselines by at least 11.6 percentage points in accuracy and 0.12 in F1. The gap is primarily driven by the use of large pretrained transformers for both modalities instead of traditional feature extraction approaches.

D. Conflict Detection

Out of 935 test samples, 244 (26.1%) show disagreement between the text-only and image-only predictions, yielding a Consistency Score of 0.739. When the two streams agree, accuracy is 94.07%. When they disagree, it drops to 69.67%—a gap of over 24 percentage points. This confirms that text-image conflicts are a genuine source of difficulty and that detecting them matters.

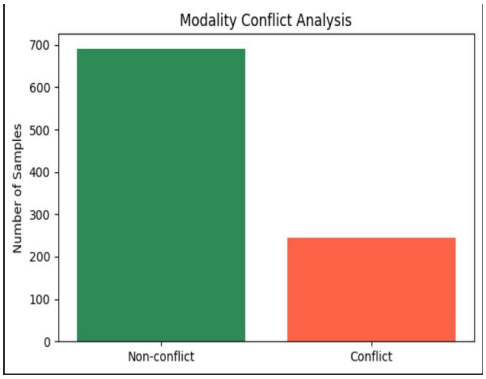


Fig. 6. Non-conflicting vs. conflicting samples in the test set.

VI. CONCLUSION

We built a multimodal sentiment analysis system for food reviews that combines RoBERTa and ViT through a late-fusion architecture. The result is a model that achieves 87.70% accuracy and F1=0.8744 on the Yelp food dataset, well above all text-only and prior multimodal baselines. More importantly, our Consistency Metric shows that when text and image agree, the model is nearly 94% accurate—but when they conflict, accuracy drops to under 70%. This 24-point gap is the clearest evidence yet that ignoring cross-modal conflict is a real problem in review analysis. Future work will explore adaptive fusion strategies that can dynamically weight the two streams depending on how trustworthy each signal is.

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